PCB Design Report

ExpressLRS ESP32 RF Development PCB (Round 1, Problem 3)

Submitted By-

Name: M. Shoyab Sadik

University: **IUBAT**

EDA Tool: **EasyEDA**

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1. Introduction

This report describes the design of an ESP32-based RF development and control PCB. The board carries an ESP32-PICO-KIT, headers for two radio modules, a programming header and power and control connectors. RF-aware placement and grounding practices are applied throughout.

2. Component List (BOM)

Table 1 lists all components with designator, value, package and source. Parts from the EasyEDA library are marked "Library"; parts built from datasheet dimensions are marked "Custom".

Designator	Component	Value	Package	Source
1	ESP32-PICO-KIT development board	-	[2 x 20-pin socket]	[Custom]
U10, U11	20-pin microcontroller header	2.54 mm	[1x20 female]	[Library]
U1, U2	10-pin header (control signals)	2.54 mm	[1x10]	[Library]
U3, U4	10-pin header (radio module 1)	2.54 mm	[1x10]	[Library]
U5, U6	10-pin header (radio module 2)	2.54 mm	[1x10]	[Library]
U7, U8, U9	3-pin connector	2.54 mm	[1x3]	[Library]
H1	1x5 female header	2.54 mm	[1x5]	[Library]
R1, R2	Resistor (radio_miso series)	150R	0402	[Library]
[test points]	Added test points	-	[package]	[Library]

Table 1: Bill of materials

3. Schematic

The schematic has four parts: the 3-pin power and control connectors (U7 to U9), the radio headers with series resistors (U3 to U6, R1, R2), the ESP32-PICO-KIT, and the control header and programming header (U1, U2, H1). Net labels such as radio_sck, radio_mosi and radio_miso connect all parts, so one name always means one net.

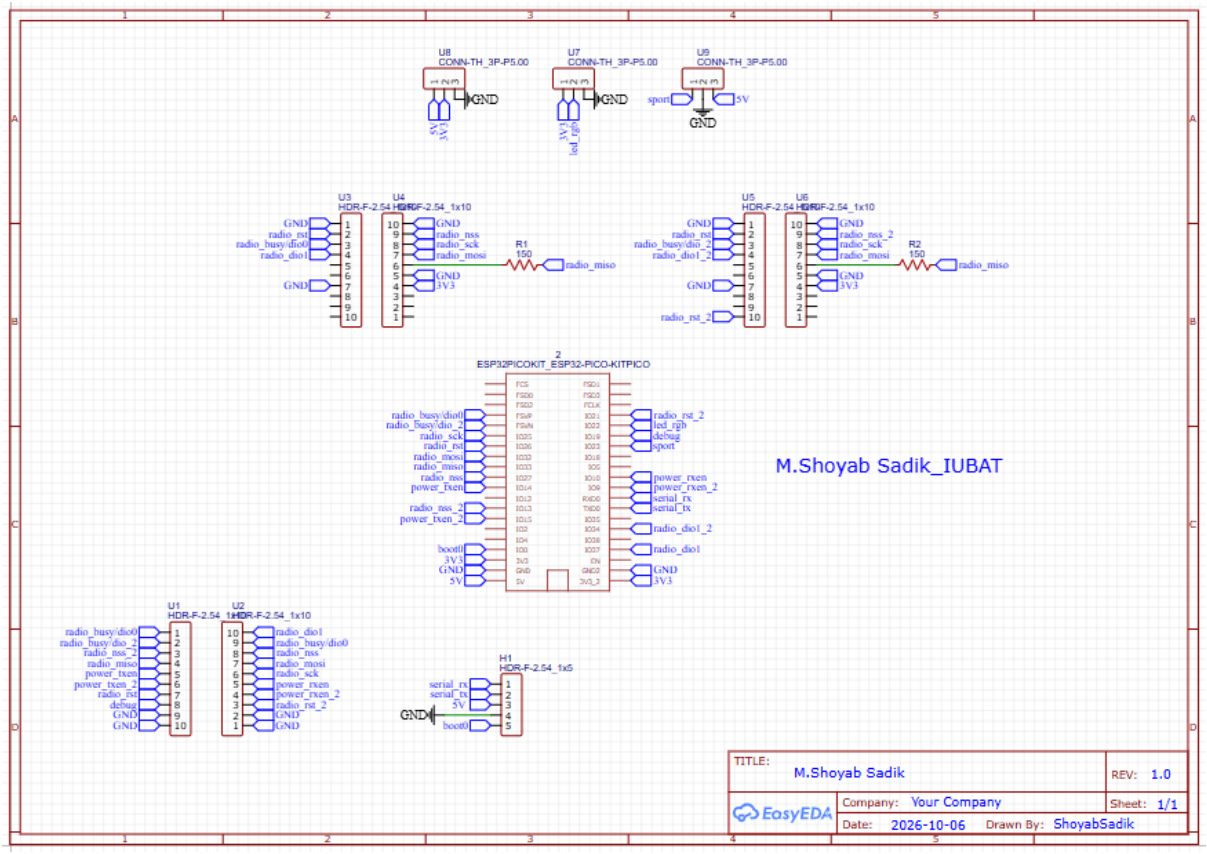


Fig1: Schematic of PCB

4. PCB Design Decisions

The board is [width: 60.452 x height: 62.156] mm with 2 layers. The ESP32-PICO-KIT is placed so its antenna end sits at the board edge, with a keep-out area that has no copper or traces on any layer. The radio headers and series resistors R1 and R2 are placed close together to keep RF and SPI paths short. Major connectors sit along the perimeter, and noisy power traces stay out of the RF section.

Requirement	How it was met
RF components and paths compact	[distance between kit, U3 to U6, R1, R2]
No noisy or high-current traces through RF section	[how power was routed]
Continuous ground reference	[GND pour on both layers, stitching vias]
Decoupling close to supply pins	[state decoupling parts added, if any, and where]
Programming and debug accessible	[H1 position and labels]
Major connectors on perimeter	[U7 to U9, H1, USB end position]
Antenna keep-out	[size and position of keep-out]
Net naming, test points, silkscreen	[test points added and labels used]

Table 3: Requirements checklist

5. Design Rules

Signal traces are 0.30 mm. Clearance is 0.3 mm and vias are 0.35 mm. A GND copper pour is used on both layers with stitching vias, and the SPI lines run over continuous ground.

6. PCB Layout Views

Figure1 shows the PCB Design view and Figure 2, 3, 4 show the 2D top and bottom layer and the 3D View respectively.

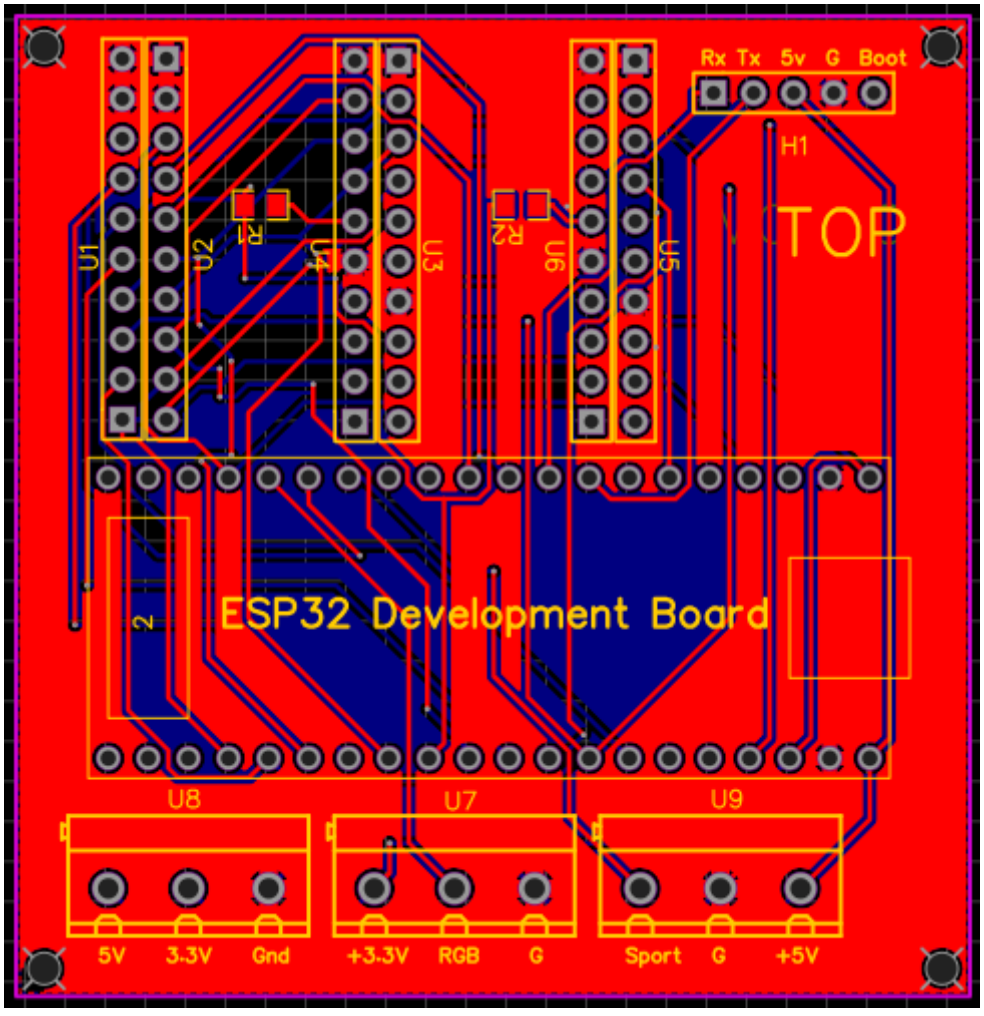


Fig2: PCB Design View

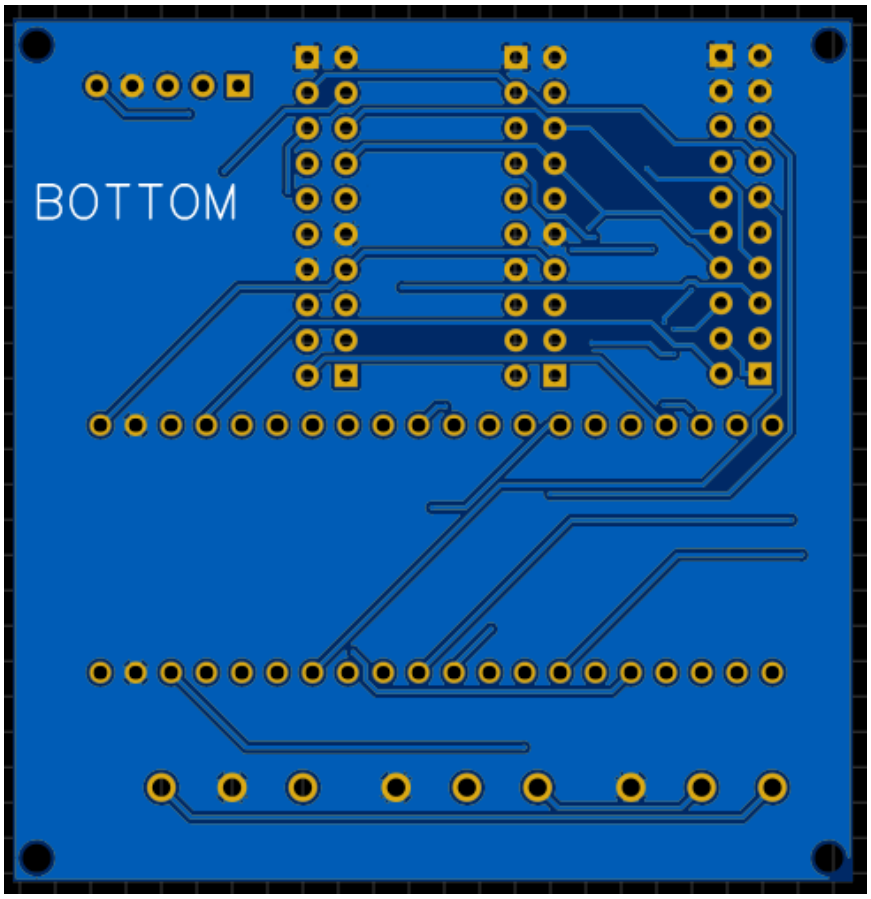


Fig3: Bottom View of PCB

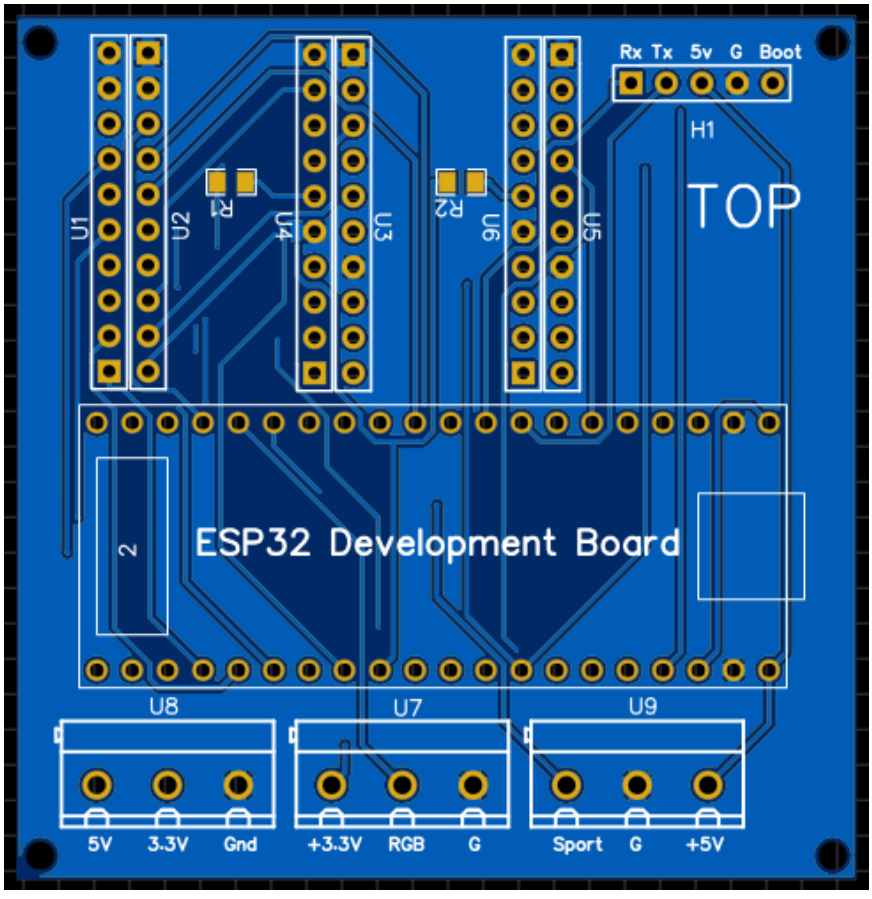


Fig4: Top View of PCB

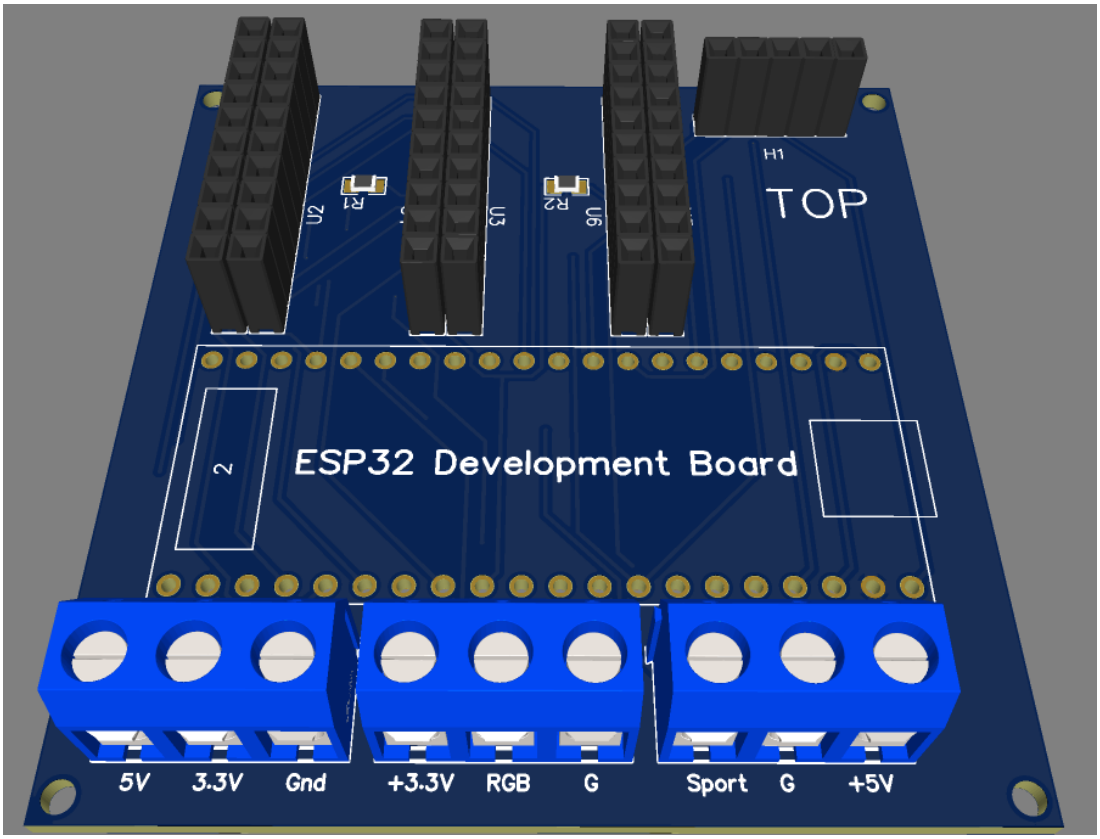


Fig5: 3D View of PCB

7. Conclusion

[Edit to match the real result:] The design meets the stated requirements: compact RF section, antenna keep-out, continuous ground, accessible programming header, perimeter connectors, clear silkscreen and a clean DRC.

8. Files Submitted

- Shoyab_Sadik_IUBAT schematic file
- Shoyab_Sadik_IUBAT board file
- Shoyab_Sadik_IUBAT pdf (all layers, one per page)
- Shoyab_Sadik_IUBAT report (this document)